

HOMEWORK 6

1. Prove that every finitely generated projective module over a PID is free.
2. Let $R = \mathbb{Z}[i]$ be the ring of Gauss numbers. Find the elementary divisors of the R -module $R/5R$.
3. Let p be prime element in a PID R , let M be a finitely generated p -primary R -module and $k = R/P$. Prove that $\dim_k(M/pM) = \dim_k({}_pM)$.
4. Find the invariant factors of the factor group $\mathbb{Z}^3/N$, where N is generated by $(-4, 4, 2)$, $(16, -4, -8)$ and $(8, 4, 2)$.
5. Find the rational canonical form of the linear operator in $\mathbb{R}^3$ given by the matrix

$$\begin{pmatrix} -2 & 0 & 0 \\ -1 & -4 & -1 \\ 2 & 4 & 0 \end{pmatrix}$$

6. Classify all finitely generated modules over $\mathbb{Z}/n\mathbb{Z}$.
7. Classify all finitely generated modules over $S^{-1}\mathbb{Z}$ where $S = \{p^n, n \geq 0\}$ and $S = \mathbb{Z} \setminus p\mathbb{Z}$ (p is a prime integer).
8. Let $A : V \rightarrow V$ be a linear operator. Prove that TFAE:
 - (a) There exists a basis such that $[A]$ is diagonal;
 - (b) There exists a basis consisting of eigenvectors of A ;
 - (c) The vector space V is a direct sum of eigenspaces of A ;
 - (d) All elementary divisors of A are linear;
 - (e) All invariant factors of A are products of distinct linear polynomials;
 - (f) The minimal polynomial m_A is a product of distinct linear polynomials.
9. Let A be a nilpotent $n \times n$ -matrix (that is $A^N = 0$ for some $N > 0$). Show that the invariant factors of A are the powers of X . Prove that $A^n = 0$.
10. Prove that an $n \times n$ -matrix is similar to its transpose A^t .